Introduction of Object Oriented Programming

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Syllabus:

Module 1: Concepts of Object-Oriented Programming: Procedure Oriented Programming, Object Oriented Programming. Properties of Object-oriented models. Concepts of Classes and Objects, Nature of an object, Nature of a class, Relationships among objects and classes. Modeling techniques of classes and objects. Paradigms of Object oriented programming. Overview of abstraction. Information hiding. Polymorphism. Method Overloading. Inheritance. Method overriding. Dynamic Binding. Message Passing. Exceptions and Exception handling. Container classes. Object Oriented Design – Process, Exploration and Analysis. Benefits of Object-Oriented Programming.

Module 2: Fundamentals of Java: Basics of Java: JVM and Bytecode, Advantages of Java. Data types, variables. Control Statements. Type conversion and casting. **Implementation of Object-Oriented Programming using Java:** Java Classes and Objects, Declaring objects, Constructors. this pointer. Method and constructor overloading. Nested and Inner class. **Inheritance:** Implementation of inheritance. super pointer. Multilevel class hierarchy. Method overriding. Abstract classes. Interfaces. Packages.

Syllabus:

Module 3: Intermediate Java: Exception Handling: Fundamentals. try and catch keywords. Nested try. throw, throws, finally keywords. Java's built-in exceptions and creating custom exceptions. **Multithreaded Programming**: Basic concept of threading, Java thread model. The main thread. Creating custom thread. isAlive() and join() methods. Thread priorities. Synchronization. Suspending, Resuming and Stopping threads. The Concurrent API.

Module 4: Advanced Java: **Java IO and Applets**: Reading and Writing to Console. PrintWriter class. Reading and Writing to files. Formatting output. Automatically closing a file. Applet Fundamentals. **Generics and Lambda Expressions**: Generic class, Bounded types. Wildcard arguments. Creating generic methods. Generic Class Hierarchies. Lambda Expressions. Method references.

Books:

1. Object-Oriented Analysis and Design with Applications by Grady Booch et. al.
2. Object-Oriented Modeling and Design With UML by Blaha, Rumbaugh
3. Java: The Complete Reference 9th Edition by Herbert Schildt
4. Effective Java by Joshua Bloch

JAVA Installation

1. Java Development Kit (JDK)
2. Code Editor/ Integrated Development Environment (IDE)
    a. Visual Studio Code
    b. Intellij
    c. Eclipse
3. Java in Visual Studio Code (**Single Pack of JDK and IDE**)

Introduction

- Object-oriented programming (OOP) is a programming paradigm that uses “Objects “and their interactions to design applications.
- Java is an Object Oriented programming (OOP) language developed by **Sun Microsystems** in 1991.
- It was originally called as **Oak** by James Gosling.
- OOP tries to map code instructions with real world, making the **code short and easier** to understand.

The Java Features

- Simple
- Secure
- Portable
- Object-oriented
- Robust
- Multithreaded
- Architecture-neutral
- Interpreted

Procedure Oriented Programming (POP) & Object Oriented Programming (OOP)

POP	OOP
Program is divided into small parts called functions.	Program is divided into parts called objects.
Importance is not given to data but to functions as well as sequence of actions to be done.	Importance is given to the data rather than procedures or functions because it works as a real world.
follows Top Down approach.	OOP follows Bottom Up approach.
It does not have any access specifier.	OOP has access specifiers named Public, Private, Protected, etc.
Data can move freely from function to function in the system.	objects can move and communicate with each other through member functions.

Procedure Oriented Programming (POP) & Object Oriented Programming (OOP)

POP	OOP
Most function uses Global data for sharing that can be accessed freely from function to function in the system.	In OOP, data can not move easily from function to function,it can be kept public or private so we can control the access of data.
It does not have any proper way for hiding data so it is less secure.	OOP provides Data Hiding so provides more security.
Overloading is not possible.	In OOP, overloading is possible in the form of Function Overloading and Operator Overloading.
Example: C, VB, FORTRAN, Pascal.	Example: C++, JAVA, VB.NET, C#.NET.